CONDITIONS ON THE WHOLE FIBRE OF A LOCALLY QUASI-FINITE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 4707 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Point-in-fibre conditions. lemma-conditions-on-point-in-fibre-and-qf (carded L26744). The pointwise conditions.
- Quasi-compact fibres of presentations. Properties of Spaces, lemma quasi-compact-fibre (cross-chapter, named). The fibre decomposition.

Source reference. Original lemma 0ACA: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4497–4497. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0AC8: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4392–4392. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ABM: focused-seeds/geometry/S9-decent-conditions-on-fibre-and-qf/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-decent-conditions-on-fibre-and-qf/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is locally of finite type. Let $y \in |Y|$. Let k be a field and let $\mathrm{Spec}(k) \to Y$ be in the equivalence class defining y. Set $X_k = \mathrm{Spec}(k) \times_Y X$*

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and let $F = f^{-1}(\{y\})$ with the induced topology from $|X|$. Consider the following conditions

(1) *F is finite,*
(2) *F is a discrete topological space,*
(3) $\dim(F) = 0$,
(4) *$|X_k|$ is a finite set,*
(5) *$|X_k|$ is a discrete space,*
(6) $\dim(|X_k|) = 0$,
(7) $\dim(X_k) = 0$,
(8) *f is quasi-finite at all points of $|X|$ lying over y.*

Then we have

$$(1) \Longleftarrow (4) \underset{f \text{ decent}}{\Longrightarrow} (5) \Longleftrightarrow (6) \Longleftrightarrow (7) \Longleftrightarrow (8)$$

If Y is decent, then conditions (2) and (3) are equivalent to each other and to conditions (5), (6), (7), and (8). If Y and X are decent, then (1) implies all the other conditions.

Proof. By Lemma 0ACA conditions (5), (6), and (7) are equivalent to each other and to the condition that $X_k \to \mathrm{Spec}(k)$ is locally quasi-finite. Thus by Morphisms of Spaces, Lemma 0ABM they are also equivalent to (8). If f is decent, then X_k is a decent algebraic space and Lemma 0ACA shows that (4) implies (5).

The map $|X_k| \to F$ is surjective by Properties of Spaces, Lemma 03H4 and we see (4) $\Rightarrow$ (1).

If Y is decent, then we can pick a quasi-compact monomorphism $\mathrm{Spec}(k') \to Y$ in the equivalence class of y. In this case Lemma 0AC8 tells us that $|X_{k'}| \to F$ is a homeomorphism. Combined with the arguments given above this implies the remaining statements of the lemma; details omitted. □